

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 9: Competitive and Profit-Maximizing Firms

Notes on the Chapter

This chapter presents the standard theory of the firm or, at least, the theory for price-taking (competitive) and profit-maximizing firms that are described by a production-possibility set. The chapter deals primarily with the firm's profit-maximization problem; the cost-minimization problem is outlined only (and a bit incompletely) in a section at the end of the chapter. The basic model of the firm is presented: A firm is described by its production-possibility set Z (various properties of which are defined) and by the objective of maximizing profit. Basic properties of the profit-maximization problem and, in particular, the profit function, are given, and (quite simple) results analogous to Afriat's Theorem are presented. This includes a discussion of the recession-cone property as necessary and sufficient (with Z closed) for existence of a solution to the profit-maximization problem for every strictly positive price vector p .

Then, beginning in Section 9.5, the techniques of *duality* in this context are presented: A candidate profit function (finite valued for all strictly positive p) is indeed a profit function if and only if it is convex and homogeneous of degree 1, two production-possibility sets have the same profit function if they have the same closure-of-their-free-disposal-convex-hulls, and this last statement is almost "if and only if," except for when both production-possibility sets give infinite profit for all strictly positive prices. This is related to the general theory of duality between convex sets and their support functions. And then differentiability of the profit function is discussed: Hotelling's Lemma is proved, along with the "converse" (that π is differentiable wherever the profit-maximization problem has a unique solution). And, finally, the cost-minimization problem is introduced and results are given (but not proved in the text; proofs are given in the *Student's Guide*.)

That is a lot of material, even for very good students, and I've found that it normally takes two class sessions to do justice to it: The first session is spent on the preliminaries (material through Section 9.4), and then an introduction to duality (talking about different production-possibility sets that give the same profit function). And

in the second session, duality is tackled. Since this is all nicely wrapped up by the end, and it will become more complicated in Chapter 10 when we return to the theory of the consumer, I stress the details of proofs in this session, emphasizing how the Separating-Hyperplane Theorem is used. I also spend time on the details of showing why uniqueness of solution implies differentiability of π ; I can then indulge in a bit of arm-waving when this result returns in the context of differentiability of the expenditure and indirect utility functions. (And this two-session lesson plan is wholly without any discussion of the cost-minimization problem.)

Notes on the Problems

Some of the problems in this chapter are quite difficult and should be assigned with care. This applies in particular to part c of Problem 10 (see the writeup below for why) and Problems 13 and 15. Note that I don't give a solution to Problem 15 either here or in the *Guide*; it is so similar to the dual consumer's problem, that there is no point in doing it until after Chapter 10 is done. Indeed (and as long as you stick to cost minimization in this context, that is, with one output and a production function), you might challenge students simply to adapt all of Chapter 10 to this alternative context; the mathematics is, of course, identical.

Problem 8 is a good exercise for testing whether students get the basic duality argument: in one direction the connection is obvious; the second clearly calls for an application of the Separating-Hyperplane Theorem but, as is indicated in the writeup following, nailing things down requires care.

Especially if you take two sessions to cover this chapter, Problem 1 should be handled with some care. Part a is a very useful exercise for the basics: It provides a concrete example of constructing π from Z . Part b, on the other hand, clearly belongs to the duality part of the chapter. One possibility would be to assign problem 1a on the first day and ask, "What would happen to π if we replaced the Z given there with all the points that lie 'below' the convex hull of the three points?" Then, on the second day, go back to part b, which in any case is a good (and simple) way to be sure that students have grasped what is being shown in Propositions 9.14 and 9.17.

Solutions to Non-Starred Problems

■ 9.1. (a) The set Z consists of three points. Therefore, π is found by finding, for each $p \in R_{++}^2$, which of the three maximizes profit. Since we are looking at strictly positive prices, and since π is homogeneous of degree 1 in p , it is enough to characterize π for p taking the form $(p_1, 1)$, where p_1 varies in the open interval $(0, \infty)$. Then for any $p = (p_1, p_2)$, $\pi(p) = p_2 \pi(p_1/p_2, 1)$.

So take $p = (p_1, 1)$. The point $(0,0)$ gives profit 0. $(-3, 2)$ gives $2 - 3p_1$. And $(-1, 1)$ gives $1 - p_1$. For $p_1 \geq 1$, $(0, 0)$ is profit maximizing (at $p_1 = 1$, so is $(-1, 1)$), and

profit is zero. For $1/2 \leq p_1 < 1$, $(-1, 1)$ is profit maximizing (and $\pi = 1 - p_1$), and for $0 < p_1 \leq 1/2$, $(-3, 2)$ is profit maximizing. So we have

$$\pi((p_1, 1)) = \begin{cases} 0, & \text{for } p_1 \geq 1, \\ 1 - p_1, & \text{for } 1/2 \leq p_1 < 1, \text{ and} \\ 2 - 3p_1, & \text{for } 0 < p_1 < 1/2. \end{cases}$$

This extends to all $(p_1, p_2) \in R_{++}^2$ as follows:

$$\pi((p_1, p_2)) = \begin{cases} 0, & \text{for } p_1 \geq p_2, \\ p_2 - p_1, & \text{for } p_2/2 \leq p_1 < p_2, \text{ and} \\ 2p_2 - 3p_1, & \text{for } 0 < p_1 < p_2/2. \end{cases}$$

This function has unique subgradients for "most" values of (p_1, p_2) . But where $p_1 = p_2/2$ and for $p_1 = p_2$, the function is "kinked" and it has many subgradients. Specifically, the subgradients of π at (p_1, p_2) are

$$\begin{cases} \{(0, 0)\}, & \text{for } p_1 > p_2, \\ \{a(0, 0) + (1 - a)(-1, 1); a \in [0, 1]\}, & \text{for } p_1 = p_2, \\ \{(-1, 1)\}, & \text{for } p_2/2 < p_1 < p_2, \\ \{a(-1, 1) + (1 - a)(-3, 2); a \in [0, 1]\}, & \text{for } p_2/2 = p_1, \text{ and} \\ \{(-3, 2)\}, & \text{for } 0 < p_1 < p_2/2. \end{cases}$$

In other words, away from the anomalous values where π is kinked, the unique subgradient is the unique profit-maximizing plan; at those values of p , we have all convex combinations of the two production plans that are (equally) profit-maximizing. And while the nontrivial convex combinations are subgradients, they aren't "profit-maximizing," because they aren't in the original Z .

But (and this is the "sort of" part), if we added those convex combinations, it wouldn't change π at all. More than that, if we added any convex combination of things in Z , or any point that lies south-west (below) a convex combination of things in Z , it wouldn't change Z . That's what Proposition 9.16 tells us. And the particular convex combinations that are subgradients of π at a particular p would be equally profit-maximizing (equal to the profit-maximizing plans already in Z). So they are "sort-of" profit-maximizing plans, at the prices for which they are subgradients.

(b) The direction $Z \subseteq Z^*$ is obvious. Every $z \in Z$ (obviously) satisfies the defining characteristic of Z^* , namely $p \cdot z \leq \pi(p)$ for all $p \in R_{++}^k$.

But Propositions 9.14 and 9.17 essentially gives us both directions at once: We know that π is convex and homogeneous of degree 1, of course, and Proposition 9.14 tells

us that Z^* is closed, convex, and has free disposal, and it generates the profit function π . Proposition 9.17 then tells us that, since both Z and Z^* are closed and convex and have free disposal, if they are not identical, either they generate different profit functions or they both generate $\pi \equiv \infty$; since they generate the same profit function $\pi < \infty$, they must be the same set. Finally, if Z^{**} is the set of all subgradients of π (ranging over all $p \in R_{++}^k$), we know that the profit function generated from Z^{**} is π (for each p , there is some $z \in Z^{**}$ with $p \cdot z = \pi(p)$, and $p \cdot z' \leq \pi(p)$ for all other z'), so applying Proposition 9.14 again tells us that $(Z^{**}) = Z = Z^*$.

The connection to part a is: In part a, we found the profit function π for a Z that is not convex and doesn't have free disposal. But if we took the free-disposal convex hull of that Z (which is closed), we'd get the same π . And the corollary tells us that this free-disposal convex hull consists of all z such that $p \cdot z \leq \pi(p)$ for all $p \in R_{++}^k$, which (of course) contains *all* the subgradients of π . If, instead, we began with the original Z , constructed π , then took all the subgradients of this π , and then constructed the free-disposal convex hull (of all the subgradients), we'd wind up in precisely the same place.

■ 9.3. The assumptions here are that Z is closed (and, of course, nonempty) and that for all p in some open neighborhood of a price p^0 (within R_{++}^k), the problem of maximizing $p \cdot z$ for $z \in Z$ has a finite supremum. For the balance of this problem let P be this open neighborhood.

Following the spirit of the analysis in the chapter, fix some (arbitrary) $z^0 \in Z$ and, for each $p \in P$, define $A(p) := Z$ and $B(p) := \{z \in Z : p \cdot z \geq p \cdot z^0\}$. It is clear that conditions a and b of the proposition hold; the key is to verify that, for this definition of the B -correspondence, condition c holds. Obviously, for each $p \in P$, $B(p) \subseteq A(p)$, the set of profit-maximizers at each p (if any exist) form a subset of $B(p)$, and $\sup\{p \cdot z : z \in A(p)\} = \sup\{p \cdot z : z \in B(p)\}$. So all the action is in proving that, for $p \in P$, $p \Rightarrow B(p)$ is upper semi-continuous and locally bounded. Proving upper semi-continuity is not difficult: Suppose $p^n \rightarrow p$ (inside P), $z^n \in B(p^n)$, and $z^n \rightarrow z$. Since each $z^n \in Z$ and Z is closed, $z = \lim_n z^n \in Z$. And since $p^n \cdot z^n \geq p^n \cdot z^0$ for each n and the dot product is continuous, $p \cdot z \geq p \cdot z^0$. Therefore, $z \in B(p)$.

As for local boundedness, suppose $p \Rightarrow B(p)$ is not locally bounded at some $p^* \in P$. Then we can produce a sequence $\{p^n\}$ with limit p^* and, for each n , a $z^n \in B(p^n)$ such that $\lim_n \|z^n\| = \infty$. Let $\hat{z}^n = z^n / \|z^n\|$, so that $\hat{z}^n$ lies on the unit circle in R^k ; by extracting a subsequence if necessary, we can find a z^* also on the unit circle such that $\lim_n \hat{z}^n = z^*$. Since $p^n \cdot z^n \geq p^n \cdot z^0$ for each n , $p^n \cdot \hat{z}^n \geq p^n \cdot (z^0 / \|z^n\|)$. And since $\lim_n p^n \cdot \hat{z}^n = p^* \cdot z^*$ while $\lim_n p^n \cdot (z^0 / \|z^n\|) = p^* \cdot z^0 [\lim_n 1 / \|z^n\|] = 0$, we conclude that $p^* \cdot z^* \geq 0$. Since z^* cannot be identically zero and p^* is strictly positive, there must be some component i of z^* that is strictly positive. And since $p^* \in P$ and P is an open set, we can find a price p^{**} near p such that $p^{**} \cdot z^* > 0$. (If $p^* \cdot z^* > 0$, we can take $p^{**} = p^*$. Otherwise, we may need to perturb p^* slightly.) But then $p^{**} \cdot \hat{z}^n$ is, for all large enough n , strictly positive and uniformly bounded away from zero, which

implies that $p^{**} \cdot z^n = \|z^n\| [p^{**} \cdot \hat{z}^n]$ grows without bound, contradicting the assumption that π is finite valued for all $p \in P$. $p \Rightarrow B(p)$ must be locally bounded, and Berge's Theorem applies at all $p \in P$.

■ 9.5. (a) Suppose that $Z = \{(0, 0), (1, -1), (2, -2), (3, -3), \dots\}$. Z is obviously additive, but it doesn't have increasing returns to scale. Suppose that $Z = \{(x_1, x_2) : \text{either } 2|x_2| \geq x_1 \geq 0 \geq x_2 \text{ or } 2|x_1| \geq x_2 \geq 0 \geq x_1\}$. This Z has increasing returns to scale but is not additive. (Draw the picture, if you aren't sure what is going on here.)

(b) Suppose Z is additive. Then either $\pi(p)$ (for this Z) is either less or equal to zero or is $+\infty$. The proof is precisely the proof given for Proposition 9.1: if $\pi(p) > 0$, then for some $z \in Z$, $p \cdot z > 0$, but then if Z is additive, for every integer n , $nz \in Z$, and $p \cdot nz = n(p \cdot z)$ which approaches ∞ as n goes to ∞ .

Therefore, if we are looking at a profit function which is either ≤ 0 or ∞ , it could be the profit function of an additive Z , or of Z that has increasing returns to scale, two classes of Z 's that have nontrivial symmetric differences.

But suppose π is the profit function of some Z and π has this property; that is, for every $p \in R_{++}^k$, $\pi(p) \leq 0$ or $\pi(p) = \infty$. Construct from Z two other production possibility sets, Z' and Z'' , as follows:

$$Z' = \{z' \in R^k : z' = \alpha z \text{ for some } z \in Z \text{ and } \alpha \geq 1\}; \text{ and}$$

$$Z'' = \{z'' \in R^k : z'' = \sum_{i=1}^{\ell} m_i z_i, \text{ for some positive integer } \ell \text{ and,} \\ \text{for each } i = 1, \dots, \ell, \text{ a positive integer } m_i \text{ and a } z_i \in Z\}.$$

In words Z' is the smallest production possibility set that contains Z and has increasing returns to scale, and Z'' is the smallest production possibility set that contains Z and is additive. (You have to prove all these claims.) Then Z' and Z'' both have the same profit function π as does Z . Which shows that the characterization we've given of profit functions of Z that have increasing returns and/or that are additive is a sharp characterization as there can be.

■ 9.6. The condition in Proposition 9.10 is that $p^n \cdot z^n \geq p^n \cdot z^m$ for all $n, m = 1, \dots, N$. But if, for $s > t$ (and I've changed from m and n to s and t to suggest time indices), z^s is not necessarily available when z^t is chosen (at prices p^t), then we can have $p^t \cdot z^s > p^t \cdot z^t$ without invalidating the model. Hence the necessary and sufficient condition is that

$$p^t \cdot z^t \geq p^t \cdot z^s \text{ for all } t \leq T \text{ and all } s < t.$$

The proof is quite simple and so is omitted.

■ 9.8. One direction easy. If $Z \subseteq Z'$, then $\pi'(p) \geq \pi(p)$ for all p . This is so because $\pi'(p) = \sup\{p \cdot z : z \in Z'\}$, and taking a supremum over a larger set must give a (weakly) larger answer.

But the other direction is less clean. For one (probably fairly obvious) thing, and supposing $\pi' \geq \pi$, Z could fail to be a subset of Z' because Z' lacks points that are never candidate solutions for the firm's maximization problem. For instance, Take any Z that contains the origin and the entire negative orthant, and let Z' be Z less the negative orthant (but with the origin still included). As long as $0 \in Z'$, no point in the (rest of the) negative orthant will ever be a solution to the firm's maximization problem, so deleting these points has no impact on the firm's profit function. Similar remarks apply to point in Z that can be written as a convex combination of other points in Z —as long as those other points remain in Z' , the absence of the convex combination doesn't affect π' —and points less than such convex combinations.

So if we want to move from $\pi' \geq \pi$ to a statement about set inclusion involving Z and Z' , we must be dealing with the closed free-disposal convex hulls of the sets that originally generated π and π' , or simply assume that those sets are themselves closed, convex, and have free disposal.

But there is a second problem. Even if Z and Z' are $\overline{\text{FDCH}}$ sets, having profit functions that are infinite-valued for all (strictly positive) p causes problems. Moreover, the problems caused by everywhere-infinite-valued profit functions are not what you would necessarily expect. We know from examples produced for Problem 9.7 (and given in the *Student's Guide*) that we could have $\pi' \equiv \pi \equiv \infty$, which means $\pi' \geq \pi$, and yet $Z' \not\supseteq Z$. But even if one of the two profit functions is identically ∞ , problems are possible. For instance, suppose that $Z = \{z \in R^k : z \leq 0\}$ and $Z' = \{z \in R^k : z_1 \leq -1\}$. In words, Z' admits any vector, as long as the first component is -1 or less. Of course, both Z and Z' are closed and convex and have free disposal. Obviously, $Z' \not\supseteq Z$. But $\pi' > \pi$, and rather dramatically so, since $\pi \equiv 0$ and $\pi' \equiv \infty$.

Here is a positive result: *Suppose Z and Z' are both closed and convex and have free disposal. If $\pi'(p) \geq \pi(p)$ for all $p \in R_{++}^k$ and $\pi' \not\equiv \infty$, then $Z \subseteq Z'$.*

I will prove this by proving its contrapositive: Suppose that $\pi' \not\equiv \infty$ and that $Z \not\subseteq Z'$. I must show that $\pi' \not\geq \pi$. Let p^0 be a strictly positive price for which π' is finite, and let z^0 be a point in $Z \setminus Z'$. Then there is a strictly separating hyperplane between the compact and convex set $\{z^0\}$ and the closed and convex set Z' . That is, for some $q \in R^k \setminus \{0\}$, $q \cdot z^0 > \sup\{q \cdot z' : z' \in Z'\}$.

I assert that $q \geq 0$: if some component of q , say component i , is strictly negative, then we can use free-disposal of Z' (and the maintained assumption that Z' is non-empty) to produce elements $z^n \in Z'$ whose components in coordinates other than i stay fixed, and whose i th component approaches $-\infty$. But then $\lim_n q \cdot z^n = \infty$ (the contribution to the inner product of all coordinates except i is fixed, and the contribution of the i th coordinate approaches ∞), violating the asserted separation.

Because of the strict inequality in $q \cdot z^0 > \sup\{q \cdot z' : z' \in Z'\}$, and because $\pi'(p^0) = \sup\{p^0 \cdot z' : z' \in Z'\} < \infty$, we can find an $\epsilon > 0$ but small enough so that

$$(q + \epsilon p^0) \cdot z^0 > \sup\{q \cdot z' : z' \in Z'\} + \epsilon \pi'(p) \geq \sup\{(q + \epsilon p^0) \cdot z' : z' \in Z'\}.$$

Having found this ϵ , let $p^* = q + \epsilon p^0$. Then

$$\pi(p^*) \geq p^* \cdot z^0 = (q + \epsilon p^0) \cdot z^0 > \sup_{z' \in Z'} (q + \epsilon p^0) \cdot z' = \pi'(p^*).$$

That is, $\pi(p^*) > \pi'(p^*)$, and hence $\pi' \not\geq \pi$.

■ 9.10. (a) Since Z is assumed to be nonempty, $\tilde{\pi}_Z(p)$ is either finite valued or equal to $+\infty$ for each p . It is also evident that $\tilde{\pi}_Z(0) = 0$. Take any $p \neq 0$: By convention, for $\alpha = 0$, $\tilde{\pi}_Z(\alpha p) = \tilde{\pi}_Z(0) = 0 = \alpha \tilde{\pi}_Z(p)$. So, to show homogeneity, we are left with the case of $p \neq 0$ and $\alpha > 0$: Let $\{z^n\}$ be a sequence out of Z such that $\lim_n p \cdot z^n = \tilde{\pi}_Z(p)$. Then $\lim_n (\alpha p) \cdot z^n = \alpha \lim_n p \cdot z^n = \alpha \tilde{\pi}_Z(p)$, and since $\tilde{\pi}_Z(\alpha p) \geq (\alpha p) \cdot z^n$ for each n , $\tilde{\pi}_Z(\alpha p) \geq \alpha \tilde{\pi}_Z(p)$. But now replace p with αp and α with $1/\alpha$ to get the reverse inequality, and $\tilde{\pi}_Z(\alpha p) = \alpha \tilde{\pi}_Z(p)$; $\tilde{\pi}_Z$ is homogeneous of degree 1.

Now take any p, p' , and $\alpha \in [0, 1]$, and let $p'' = \alpha p + (1 - \alpha)p'$. Let $\{z^n\}$ be a sequence out of Z such that $\lim_n p'' \cdot z^n = \tilde{\pi}_Z(p'')$. Then

$$\tilde{\pi}_Z(p'') = \lim_n p'' \cdot z^n = \lim_n (\alpha p + (1 - \alpha)p') \cdot z^n = \lim_n [\alpha(p \cdot z^n) + (1 - \alpha)(p' \cdot z^n)].$$

But $\tilde{\pi}_Z(p) \geq p \cdot z^n$ and $\tilde{\pi}_Z(p') \geq p' \cdot z^n$ for each n , so from the line just above, we get

$$\tilde{\pi}_Z(p'') \leq \lim_n [\alpha \tilde{\pi}_Z(p) + (1 - \alpha) \tilde{\pi}_Z(p')] = \alpha \tilde{\pi}_Z(p) + (1 - \alpha) \tilde{\pi}_Z(p').$$

We have to be a bit careful in this when dealing with cases such as $\alpha = 0$ and $\tilde{\pi}_Z(p) = \infty$, but the conventions we've established (and the direction of the inequality) make it clear that the inequality holds, which shows that $\tilde{\pi}_Z$ is convex.

(b) Suppose $x, x' \in X^*$; that is, $f(x) < \infty$ and $f(x') < \infty$. Then by convexity of f , for all $\alpha \in [0, 1]$, $f(\alpha x + (1 - \alpha)x') \leq \alpha f(x) + (1 - \alpha)f(x') < \infty$, and $\alpha x + (1 - \alpha)x' \in X^*$; the set X^* is convex. And for all $x \in X^*$ and $\alpha \geq 0$, $f(\alpha x) = \alpha f(x) < \infty$, so $\alpha x \in X^*$. Therefore, X^* is a cone.

(c) We begin by giving $\tilde{\pi}_Z$ for the two examples suggested in the problem:

If Z is the closed non-positive orthant, or R_-^k , then for any $p \in R_+^k$, $p \cdot z \leq 0$ for all $z \in Z$, hence $\tilde{\pi}_Z(p) = 0$ (for $p \in R_+^k$). And for any $p \notin R_+^k$, p has some component which is strictly negative, say the i th. Let $z^n = (0, 0, \dots, 0, -n, 0, \dots, 0)$, where the $-n$ appears in the i th component position; this z^n is in Z , and $p \cdot z^n = -np_i$, which

approaches $+\infty$ as $n \rightarrow \infty$, so $\tilde{\pi}_Z(z) = \infty$. So for this Z , $P_Z^* = R_+^k$, for which $\tilde{\pi}_Z$ is identically 0.

In the second example, Z contains the closed non-positive orthant R_-^k but, in addition, it contains points (x_1, x_2) where $x_1 < 0$ and $x_2 \leq |x_1|^{1/2}$. Since Z has gotten larger $\tilde{\pi}_Z(p)$ in this case can only grow relative to the first example; in particular, $\tilde{\pi}_Z(p) = \infty$ for all $p \notin R_+^k$. And now, in addition, $\tilde{\pi}_Z(p) = \infty$ for all p of the form $(0, p_2)$, for $p_2 > 0$. For all the other $p \in R_+^k$, $\tilde{\pi}_Z(p) < \infty$; that is, $P_Z^* = \{(p_1, p_2) : p_1 > 0, p_2 \geq 0\} \cup \{(0, 0)\}$. For $p = (0, 0)$, we obviously have $\tilde{\pi}_Z(p) = 0$, so this leaves $p = (p_1, p_2)$ where $p_1 > 0$ and $p_2 \geq 0$. Fix such a p . It is clear that to maximize $p \cdot z$, we take $x_2 = |x_1|^{1/2}$. So letting the variable x be $|x_1|$, the objective is to maximize $p_2 x^{1/2} - p_1 x$ in x , which has first-order condition $p_1 = p_2/(2x^{1/2})$, or $x = (p_2/2p_1)^2$, so that

$$\tilde{\pi}_Z(p) = p_2 \frac{p_2}{2p_1} - p_1 \frac{p_2^2}{4p_1^2} = \frac{p_2^2}{4p_1}.$$

Now to tackle general continuity properties. Obviously, $\tilde{\pi}_Z$ is always continuous on the complement of P_Z^* , since it is identically infinite there. And it is continuous on the interior of P_Z^* , since it is convex, and (finite-valued) convex functions are always continuous on the interior of their domains. So all we need to worry about is continuity along and approaching the boundary of P_Z^* .

The two examples tell us some things about how $\tilde{\pi}_Z$ can behave along and near the boundary. The first example shows that it can be finite valued on the boundary: That is, P_Z^* can be closed. But the second example shows that it is also possible that $\tilde{\pi}_Z$ can be infinite along at least part of the boundary of P_Z^* . These examples also show that, as you approach a point on the boundary of P_Z^* from outside $P^*(Z)$, $\tilde{\pi}_Z$ fails to be continuous: In the first example, for instance, along the sequence of points $p^n = (-1/n, 1)$, $\tilde{\pi}_Z(p^n) = \infty$, but at the limit point $(0, 1)$ its value is 0.

So we will not have full continuity. We do, however, get lower semi-continuity, by which is meant: if $p^n \rightarrow p$, then $\liminf_n \tilde{\pi}_Z(p^n) \geq \tilde{\pi}_Z(p)$, where the $\liminf$ is allowed to be $+\infty$. Note that we are not restricting the p^n or p to lie in P_Z^* . To see this, fix some sequence $\{p^n\}$ with limit p . Let $\{z^m\}$ be a sequence from Z such that $\lim_m p \cdot z^m = \tilde{\pi}_Z(p)$. Then for each n and m , $\tilde{\pi}_Z(p^n) \geq p^n \cdot z^m$. Therefore, for each m , $\liminf_n \tilde{\pi}_Z(p^n) \geq \lim_n p^n \cdot z^m = p \cdot z^m$. But the term on the right-hand side of this inequality is constant in m , so it is greater or equal to the limit of the right-side side as m goes to ∞ , which is $\liminf_n \tilde{\pi}_Z(p^n) \geq \tilde{\pi}_Z(p)$.

Note that lower semi-continuity implies: If p is on the boundary of P_Z^* but is not in P_Z^* (that is, $\tilde{\pi}_Z(p) = \infty$), then along any sequence $\{p^n\}$ with limit p , $\lim_n \tilde{\pi}_Z(p^n) = \infty$.

We might hope for a bit more: We might hope to show that $\tilde{\pi}_Z$ is continuous on P_Z^* . (That is, the only discontinuities that occur happen between P_Z^* and its complement.) But the second example shows that this fails. Let $p^n = (1/n^2, 2/n)$, so $\lim_n p^n = 0$.

Of course, $\tilde{\pi}_Z(0) = 0$, but $\tilde{\pi}_Z(p^n) = (4/n^2)/(4/n^2) = 1$. (And we can find sequences that converge to zero for which the corresponding values of $\tilde{\pi}_Z$ diverge to infinity or converge to 0.)

■ 14. Here's my result:

Suppose that, for all y and y' from Y and $\alpha \in [0, 1]$, if $x \in V(y)$ and $x' \in V(y')$, then there exists $x'' \in V(\alpha y + (1 - \alpha)y')$ such that $x'' \leq \alpha x + (1 - \alpha)x'$. Then $C(y, r)$ is convex in y , holding fixed the vector of input prices r .

Proof. Fix r , y , y' , and $\alpha \in [0, 1]$. Fix as well some $\epsilon > 0$. At these input prices, $C(y) = \inf_{x \in V(y)} r \cdot x$, so we can find some $x_\epsilon \in V(y)$ such that $r \cdot x_\epsilon - \epsilon < C(y)$. (The infimum is finite, since $V(y) \subseteq R_+^n$, and factor prices are nonnegative.) Similarly, there is some $x'_\epsilon \in V(y')$ such that $r \cdot x'_\epsilon - \epsilon < C(y')$. But then, by the assumed property, there is some $x'' \in V(\alpha y + (1 - \alpha)y')$ such that $x'' \leq \alpha x_\epsilon + (1 - \alpha)x'_\epsilon$. And then,

$$C(\alpha y + (1 - \alpha)y') \leq r \cdot x'' \leq r \cdot (\alpha x_\epsilon + (1 - \alpha)x'_\epsilon) = \alpha(r \cdot x_\epsilon) + (1 - \alpha)r \cdot x'_\epsilon$$

$$\leq \alpha(C(y) + \epsilon) + (1 - \alpha)(C(y') + \epsilon) = \alpha C(y) + (1 - \alpha)C(y') + \epsilon.$$

Since $\epsilon > 0$ is arbitrary here, this gives $C(\alpha y + (1 - \alpha)y') \leq \alpha C(y) + (1 - \alpha)C(y')$, which is, of course, convexity of the C function in y .

About this, two comments:

- You can avoid the ϵ s if you assume that all the input-requirement sets are closed, so a solution to the cost-minimization problem exists.
- It is implicit in the statement of my result that Y is convex, since I am saying that if y, y' are in Y , then for any $x \in V(y)$, $x' \in V(y')$, and α , there exists some $x'' \in V(\alpha y + (1 - \alpha)y')$. The existence of such an x'' implies $\alpha y + (1 - \alpha)y' \in Y$.

■ 15. I reiterate that I am not supplying a solution to Problem 15 here. If you change the words but not the symbols and stick to the cost-minimization problem where technology is formulated with a production function, Chapter 10 is a rather extended solution.